

Detection of Cavities from Oral Images using Convolutional Neural Networks

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Abstract— The task of automatic dental cavity detection from oral images has been recently revolutionized due to several works that uses convolutional neural networks (CNNs). One of the challenges that faces this research is the ability of these systems to perform well on images collected from varied sources. Hence, this work proposes a CNN and also studies its ability to perform on images collected from various sources. For this, a custom-made CNN called Dental-Net is proposed to automatically detect cavities in the teeth from oral photographic images. A dataset with a total of 609 diverse images consisting of teeth with and without cavities, shot from varied positions and orientations are collected from a variety of sources and by web scraping. The images from the dataset are resized, converted to grayscale, followed by normalization and on-the-fly data augmentation. The dataset is then used to train the Dental-Net model. In this study, the performance of four pre-trained models namely VGG16, MobileNetV2, InceptionV3, and ResNet18 have also been evaluated. The results show that the Dental-Net model achieves the highest accuracy of 94.25% and 91.09% on the training and the validation set respectively. Comprehensive analysis of this study reveals positive results that can be improved in the future and be implemented on a commercial scale.

Keywords—deep neural network, oral photographic images, cavity detection, convolutional neural network,

I. INTRODUCTION

Cavities are one of the most common dental problems and can occur in people of any age group. These are areas of hard surface of the teeth that are permanently damaged and may develop into small holes or openings. They can be caused by bacteria, sugary drinks or food stuck in the teeth. Not brushing the teeth well adds on to the cause. If left untreated, they expand and go deeper into the tooth causing sharp nerve pain and may even lead to infections or tooth loss. If the tooth has a high level of decay, one of the treatments for this is the “Root canal” operation. Early diagnosis of this is helpful to alert people and take preventive measures for stopping the decay of the tooth.

Usually, a dentist performs a physical examination of teeth, and reviews X-rays of a patient to deduce the presence of any cavity in the teeth. This is a time-consuming process both for a patient and the dentist. To reduce the complexity of the procedures and make the system reliable and cost efficient for patients, an automated mechanism to detect the cavity from oral photographs accelerates the process and makes it inexpensive for the diagnosis. Traditionally, machine learning (ML) methods were used for building the automated systems. However, these methods required a lot of domain expertise and human intervention for attaining good results. Nowadays deep learning (DL) methods are used for the detection and development of automated systems due to their good learning ability and low generalization error [1].

DL algorithms have an incredible number of applications in the area of medical image processing. They are effectively used in problems like object detection (mitosis and cells), segmentation (kidney segmentation, tumor segmentation, COVID-19 X-ray), classification and many others. Convolutional neural networks (CNNs) are a kind of deep learning model that perform exceptionally well in these types of problems due to its network architecture, which has been designed to learn and identify patterns in input images.

Literature shows that research has been done with respect to automatic analysis of dental diseases using CNNs [3-11]. It is equally important that such automated systems work well for images collected from a variety of sources. This is a common challenge in the field of medical imaging analysis as the images acquired from varied sources are often different from each other due to reasons like variability in the devices or operator and operating conditions. Hence, in this research we have focused on the use of CNNs for recognizing cavities from oral photographic images collected from different sources. An oral photograph is a primary imaging technique that can be used to diagnose cavities. To detect dental cavities,

we develop a novel CNN model called Dental-Net and subsequently compare its performance with existing state-of-the-art pre-trained CNN architectures.

The manuscript is organized as follows: Section II details the related work on oral images. Section III describes the sources used for obtaining the dataset. Section IV outlines the proposed method followed by implementation details in Section V. Section VI presents the results obtained and comparison experiments carried out in this research followed by conclusion in Section VII.

II. RELATED WORKS

The following paragraphs provide a review on the literature about dental cavity detection.

A methodology for the segmentation of dental caries regions based on level set using X-ray images was proposed [2]. A detection accuracy of 98% was achieved using a relatively small dataset of just 120 images. A CNN algorithm was used in detecting dental caries from periapical radiographs [3]. The proposed methodology was able to correctly identify 337 out of 340 regions, attaining accuracy of 83% at the classification phase. A mask R-CNN architecture along with fine-tuning was used for the detection and localization of tooth structures. The model gave an F1 score of 0.875 and a mean intersection over union (IoU) of 0.877 [4]. The findings were based on panoramic radiographs that were manually annotated by an oral radiologist. A dataset of dental colour and X-ray photographs was used to identify regions of dental cavities using a deep learning approach [5]. A pre-trained hybrid Mask-RCNN was used for region prediction, which attained an accuracy between 78 and 92%.

Dental cavity detection using deep learning was proposed, where a dataset of 2,417 dental colour photographs of human teeth was created and used. A CNN model was trained using transfer learning and image augmentation which achieved an accuracy of 92.5% [6]. A Sobel method for detecting edges in the dental X-ray images was used for the classification of dental cavities [7]. This along with a deep CNN showed a good accuracy of 96%. An ensemble of two CNNs was used with dental X-ray images for dental caries detection [8]. The image pre-processing method applied to each of the CNNs were different, and the final SoftMax score of both the CNNs were fused to give a prediction. This method showed good performance yielding an accuracy of 99.13%. A single shot multibox detector (SSD) was applied to dental colour photographs [9]. The model was assessed for handling both image classification and localization tasks for screening dental caries. Hard-negative mining was also used during the training process to minimize the occurrence of false positives and the system exhibited a classification area under the curve (AUC) of 85.65%. The results obtained in [10] highlighted the potential utility of CNN for detecting dental caries in periapical radiographs attaining an AUC of 0.917. Using SVM, a two-step detection scheme was proposed: "C versus

(VNC + NSC)", and "VNC versus NSC" for the detection and classification of dental caries lesions based on smartphone color images [11]. According to the results, "C versus (VNC plus NSC)" achieved an accuracy of 92.37% whereas "VNC versus NSC" achieved an accuracy of 83.33%. In a research, it was investigated whether a U-shaped CNN can improve clinicians' performance in detecting caries on bitewing radiographs [12]. In regards to diagnostic performance, the CNN performed as follows on the test dataset: precision, 63.29%; recall, 65.02%; and F1-score, 64.14%. The presence and absence of root caries was detected using ML techniques [13]. Support vector machines (SVM) obtained the highest accuracy of 97.1% for the dental images obtained from a National Health and Nutrition Examination Survey.

III. DATASET

In this research we used a dataset of 609 colored dental photographic images. The images were taken from various sources and by performing web scraping. The dataset consisted of tooth images with and without cavities taken at different angles, distances, and lighting conditions, to make the proposed Dental-Net model more robust and generalized. Fig. 1 demonstrates the variability of the teeth images in the dataset. Fig. 1(a) shows teeth images with and without cavity captured using dental cheek retractors for a full-frontal view of the teeth [17]. Fig. 1(b) depicts teeth images with and without cavity from Maxilla and Mandible (upper and lower jaw) as viewed from inside the mouth [18]. Fig. 1(c) and 1(d) represent images of teeth with and without cavities as they would appear from outside the mouth and at an acute camera angle respectively, that were obtained by running a Python-based web scraping program which would find and download images on the internet matching the required search parameter. Fig. 1(e) illustrates single tooth images which were obtained from Kaggle [16]. Table 1 shows the distribution of the images in the dataset from various sources.

IV. PROPOSED METHODOLOGY

A. Data Pre-processing and Augmentation

All the input oral photographic images in the dataset were resized to a size of 256 x 256 x 3. The images were later converted to grayscale images of size 256 x 256 x 1. Grayscale normalization was performed for reducing the effects of illumination's differences. Due to limited number of images, data augmentation techniques were applied on the

TABLE I: DISTRIBUTION OF THE IMAGES IN THE DATASET

Dataset Source Reference	Number of Images
[14]	90
[15]	200
Web Scraping	229
[16]	90



Fig. 1. Varied oral images in the dataset

dataset for avoiding overfitting and creating new augmented data. In this work, the following augmentations were used: random rotation of images by a range of 30 degrees, random zooming of some images by 20%, random increase and decrease of the brightness of some images by 15%, random flipping of images horizontally, random shifting of images in the vertical direction by 10% of height, and random shifting of images in the horizontal direction by 10% of width. The preprocessed images are then fed to the Dental-Net model.

B. The Proposed Model

This paper proposes a CNN model called Dental-Net. CNNs are used in a variety of applications that analyse and recognize images [17,18]. CNNs are more efficient and accurate due to its learning ability to capture the features existing in the images of a dataset, by using the multiple feature extraction at succeeding stages. A CNN's architecture consists of several convolution and pooling layers, a flatten layer, and several dense layers (fully connected layers). It also contains three basic components: activation function, batch normalization, and dropout. In the convolution and the pooling layers, the input images are analysed and features are extracted, while in the final fully connected layer, the classification is performed.

The architecture of the proposed Dental-Net model is illustrated in Fig. 2. It consists of 4 convolutional layers each followed by a max pooling layer. The output of the final max pooling layer is flattened and is fed to a fully connected layer and then to an output layer. All the filters in all the 4 convolutional layers are of size 3×3 . All the filters in all the max pool layer are of size 2×2 .

The first convolutional layer of the Dental-Net takes the dental photograph as input. It uses 32 convolution filters and outputs 32 feature maps of size 256×256 . The output tensor is subjected to batch normalization followed by max pooling with a stride of 2. The size of the output at this stage is $128 \times 128 \times 32$.

In the next layer of convolution, the output of the previous max pool layer is used as an input. In this layer the tensor size generated is $128 \times 128 \times 64$. This is followed by a dropout layer and batch normalization that produces the output with the same dimensions. The output is then passed to a max pool layer with a stride of 2 that reduces the image size to $64 \times 64 \times 64$.

Similarly, the third convolutional layer takes the output tensor of the previous max pool layer as an input and results in a tensor of size $64 \times 64 \times 64$. This is followed by batch normalization and max pool layer. The output of the max pool layer is of size $32 \times 32 \times 64$.

The final convolutional layer takes the input of the previous max pool layer and outputs a tensor of size $32 \times 32 \times 128$. This is followed by a dropout layer and batch normalization. The max pool layer then converts the tensor to a size of $16 \times 16 \times 128$.

The output of this max pool layer is then flattened into a single dimensional neuron of size 32768, and is then passed to a fully connected layer of 128 neurons with Relu as the activation function. This is followed by a dropout layer and a final dense layer with sigmoid as the activation function, that classifies between teeth with and without cavities. The dropout parameter is set to 0.1 for the proposed model.

V. IMPLEMENTATION DETAILS

The Dental-Net model was built using Jupyter Notebook environment with python version of 3.8.8. The proposed Dental-Net was trained for 25 epochs, with Adam's propagation and binary cross entropy as the empirical loss function [18]. It was observed that the validation loss increased after 25 epochs, which resulted in a reduction of validation accuracy. Due to this, the epochs were set to 25 to prevent the model from overfitting. The model was tested using 2 different optimizers: RMSprop and Adam's propagation. Adam's propagation produced better results in terms of accuracy and loss than RMSprop, therefore Adam's propagation was used to optimize the proposed model.

The stride parameter for the convolution layer was set as 2. The training and the validation set were split into a ratio of 80:20. Relu was used as the activation function for the convolution and fully connected layers except the final fully connected layer. Sigmoid activation function was used for the last layer of the CNN to carry out binary classification. Accuracy and loss were used as the evaluation metrics. In order to reduce the effects of illumination differences, Grayscale normalization has been used to improve the performance of the model on RGB images. Also, normalization is considered so that CNN works in a faster manner [26].

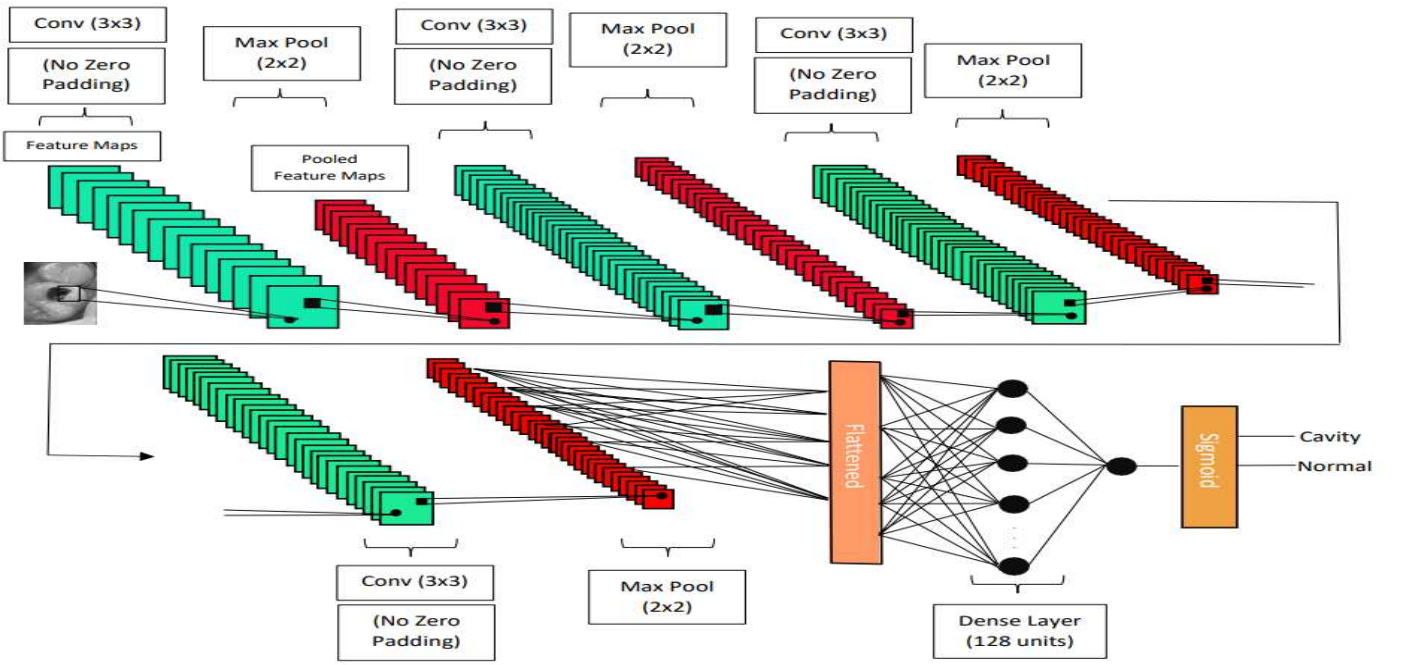


Fig. 2. The proposed architecture of Dental-Net

VI. EXPERIMENTAL RESULTS

A. Training the Dental-Net Model

The values of accuracy and loss were recorded for both the training and the validation sets for the Dental-Net model. Table 2 presents the values of the accuracy and the loss obtained for 1, 5, 10, 15 and 25 epochs. The proposed model attained an accuracy of 94.25% and 91.09% on the training and the validation set respectively after 25 epochs as shown in Table 2. The loss generated on the training and the validation sets after 25 epochs were 0.1612 and 0.3369 respectively.

B. Comparison with Pre-trained Models

In this research, the performance of the proposed Dental-Net model was compared with four popular pre-trained models that are currently being used for research and commercial purposes. They are: VGG16 [27], MobileNetV2 [28], InceptionV3 [29], and ResNet18 [30]. The accuracies and the loss generated by these pre-trained models were compared with the proposed Dental-Net model. From the results attained, it is inferred that the Dental-Net model has outperformed all the pretrained models in terms of

accuracy and loss on both the training and the validation set as shown in Table 3. It is observed that InceptionV3 is higher in accuracy gap while VGG16 is having the least accuracy gap. With regard to the number of parameters considered simultaneously, MobileNetV2 and ResNet18 have a better performance. While MobileNetV2 has the fewest parameters and second lowest accuracy in comparison with the Dental-Net model, VGG16 has the highest accuracy, consisting of the highest number of parameters. ResNet18 has obtained the second highest accuracy having second fewest parameters, whereas InceptionV3 consists of the second highest number of parameters but has attained the least accuracy in comparison to ResNet18, MobileNetV2, VGG16 and the Dental-Net model. The results obtained from Table 3 exhibit that the bias for MobileNetV2 and InceptionV3 is low as they are attaining a good accuracy on training data, but they don't perform very well on the testing set causing a high variance, that leads to overfitting. Whereas the Dental-Net model has achieved a low bias and low variance, containing only around 4.3 million parameters, demonstrating its reliability and robustness for detecting cavities in teeth.

TABLE II: ACCURACY AND LOSS OF THE DENTAL NET MODEL AT DIFFERENT EPOCHS

Epochs	Training Accuracy	Training Loss	Validation Accuracy	Validation Loss
1	0.7351	3.177	0.2523	5.1310
5	0.8522	0.3519	0.7568	0.9942
10	0.8973	0.2581	0.8378	0.7355
15	0.9076	0.2324	0.8288	0.8029
20	0.9220	0.2014	0.9279	0.1840
25	0.9425	0.1612	0.9109	0.2315

TABLE III: COMPARISON WITH OTHER PRE-TRAINED MODELS ON THE TRAINING AND VALIDATION SET

Model	Training Set Accuracy	Training Set Loss	Validation Set Accuracy	Validation Set Loss
Dental-Net	94.25±0.015%	0.1612	90.09±0.015%	0.2365
VGG-16	91.25±0.015%	0.2425	87.76±0.015%	0.1430
MobileNetV2	90.38±0.015%	0.6578	84.55±0.015%	0.2945
InceptionV3	91.50±0.015%	0.5495	82.65±0.015%	0.5977
ResNet18	90.88±0.015%	0.3162	86.78±0.015%	0.1670

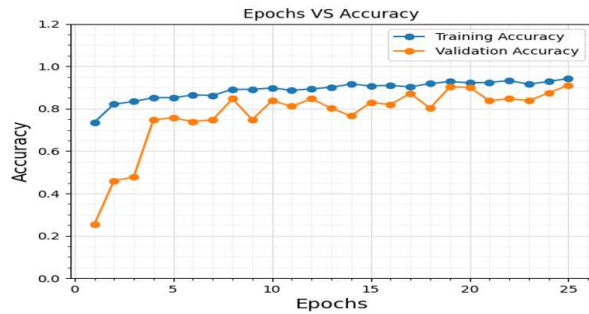


Fig. 3. Train and validation accuracy of Dental -Net for 25 epochs.

The graphs in Fig. 3 and Fig. 4 shows the training and validation accuracy and loss respectively for 25 epochs. Fig. 3 depicts the relationship between the epochs and the accuracy, while Fig. 4 illustrates the relationship between the epochs and the loss, of the Dental-Net model over 25 epochs for the training and the validation datasets.

VII. CONCLUSION

Through application of computer vision and image analysis to a catalog of images gathered from open access data repositories and web scraping, in this paper we propose Dental-Net model designed specifically for the detection of cavities in teeth. Cavities identification experiments were conducted, and the results were compared to four different models. Based on the experimental results, the Dental-Net model showed the highest of accuracy for both training and validation sets. A variety of model parameters were also analyzed and applied in order to better understand which features were critical for classifying the presence of cavities and decay in patients' teeth, that would aid clinicians in providing an accurate screening and improve certainty and transparency between patients and clinicians. The presented model can be used in future research and development applications that assist researchers and developers in designing low-cost systems and can greatly reduce the time and costs involved in commercial and expensive production processes. The proposed Dental-Net model, however, by no means is a commercially ready solution. We hope that researchers and data scientists can use the convincing results of this research to further improve the deep learning solutions for detecting cavities in oral photographs more accurately and to have a greater impact on the society.

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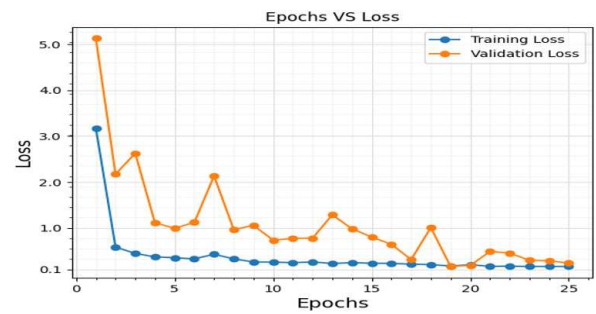


Fig. 4. Train and validation loss of Dental -Net for 25 epochs

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